Comparative Modelling of Real Estate Sales via Linear Regression and Generalised Additive Models: An Empirical Analysis and Predictive Performance Evaluation

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Summary

This empirical study evaluates the comparative predictive performance of linear regression and generalized additive models (GAMs) in residential real estate valuation using a dataset of 522 transactions. Diagnostic analyses revealed substantial violations of linear regression assumptions, including non-normality (Shapiro-Wilk $W = 0.937$, $p < 0.001$) and heteroscedasticity (Breusch-Pagan $\chi^2 = 147.25$, $p < 0.001$). While logarithmic transformation (optimal $\lambda = -0.195$) improved model fit ($R^2 = 0.834$ vs. 0.829 in linear form), residual diagnostics indicated persistent assumption violations. GAMs demonstrated superior predictive accuracy, particularly for capturing non-linear relationships, achieving lower root mean square error values (RMSE = 0.0194 for square footage vs. 0.0206 in log-linear models) and explaining 83.5% of price variance. Cross-validation results confirmed GAMs' enhanced stability (SD of $R^2 = 0.015$ vs. 0.052 for linear models). The analysis identifies square footage ($\beta = 101.8$, $p < 0.001$) and construction quality (ΔRMSE $= -0.001$) as the most influential predictors.These findings suggest GAMs should be prioritized for accurate valuation in contexts requiring predictive precision, while acknowledging linear models retain utility for their interpretability in exploratory analyses.

Keywords: *Hedonic, GAMs, Diagnostics, Prediction, Valuation, ML, Non-parametric, Cross Validation*

0.0.1 Introduction

Accurate real estate price prediction is vital for financial stability and market efficiency, impacting mortgage lending and investment. Research indicates that precise predictions mitigate financial losses and bolster social stability, as shown in studies on machine learning in real estate (Yang, 2025). This study aims to rigorously compare the predictive performance of linear regression and generalised additive models (GAMs) in modeling residential sales prices.

Research Problem

Traditional valuation approaches, particularly hedonic price models based on linear regression, have been the cornerstone of real estate analysis (Rosen, 1974). However, these models operate under strict assumptions of linearity and may fail to capture the complex, non-linear relationships that often characterize real estate markets. This can lead to inaccurate predictions and potentially flawed decision-making. The increasing availability of data and computational power allows for exploring more flexible modelling techniques.

Main Objectives

To rigorously compare and evaluate the predictive performance of linear regression and generalized additive models (GAMs) in modelling residential real estate sales prices, utilizing cross-validation techniques.

Specific Objectives

1. To develop and rigorously assess a linear regression model for predicting housing prices, emphasizing diagnostics and validation.
2. To implement and critically evaluate a GAM to predict housing prices, assessing predictive accuracy and variable significance via cross-validation.
3. To compare the strengths and weaknesses of both modelling approaches and offer evidence-based recommendations for future real estate modelling efforts.

0.0.2 Literature Review

The quantitative analysis of real estate markets has historically relied upon hedonic pricing models, wherein Rosen's (1974) framework established property value estimation via linear regression of characteristic attributes. However, the multi-factorial nature of real estate markets introduces non-linear relationships that may limit the efficacy of linear models. Empirical studies, such as Selim (2009) and Brunauer et al. (2010), have demonstrated the superior predictive performance of non-parametric and semi-parametric methodologies in such contexts. Concurrently, the proliferation of Automated Valuation

Models (AVMs), as reviewed by El Jaouhari et al. (2024), reflects the increasing adoption of statistical learning techniques for efficient property valuation. Methodologically, Khandani, Kim, and Lo (2010) emphasize the need for cross-validation and outside-sample testing for robust comparative evaluation of predictive models in real estate price forecasting.

Justification for Linear Regression and GAMs

Applying linear regression to real estate valuation provides an initial analytical framework, leveraging its inherent interpretability and established statistical properties. This approach facilitates the quantification of linear relationships between property attributes and sales prices, serving as a foundational baseline. However, the inherent constraints of linearity necessitate the exploration of alternative modelling strategies capable of capturing more complex data structures. Generalized additive models (GAMs) offer an extension to the linear regression paradigm by accommodating non-linear relationships between predictor variables and the response.

0.0.3 Methodology

Data Source

This study employs a dataset of 522 residential real estate transactions recorded in a mid-western urban area during 2002. The dataset comprises 13 variables: sales price, square footage, bedroom count, construction quality, air conditioning, highway adjacency, architectural style, Lot size, construction year and a unique identifier assigned to each transaction (Kutner et al., 2005).

Statistical Analysis

To analyze the relationship between Sales and the property characteristics, both a linear regression model and a Generalized Additive Model were employed. The property characteristics included finished square footage, number of bedrooms, number of bathrooms, air conditioning, garage capacity, presence of a pool, year of construction, construction quality, architectural style, lot size, and adjacency to a highway. Model performance assessment and selection were conducted using k-fold cross-validation. All analyses were performed using the statistical software R.

Model Specification

Regression Model

Regression is a statistical technique used to model and analyze the relationship between a dependent variable and one or more independent variables. It helps in understanding how the typical value of the dependent variable changes when any one of the independent variables is varied while the others are held fixed (Montgomery et al., 2021). The general form of a regression model:

$$y = X\beta + \varepsilon$$

Where,

- y is the response variable Sales price, which is an $n \times 1$.
- X is an $n \times p$ matrix corresponding to the predictor variables.
- β is an $n \times 1$ vector containing of the regression coefficients of the predictor variables.
- ε is an $n \times 1$ vector of the errors.

Model Assumptions

1. Normality of residuals
2. Homoscedasticity
3. Independence of residuals
4. Linearity

Model Diagnostics

Model diagnostics are crucial for assessing the validity of the linear regression model and ensuring that the assumptions underlying the model hold true. Several techniques and plots can be used to check whether the assumptions of linearity, independence, homoscedasticity, and normality are satisfied.

Residual Plot

Residual plots are used to check the assumption of linearity and homoscedasticity. The residuals 'bounce randomly' around the 0 line, suggesting the relationship is linear, and the residuals roughly form a 'horizontal band' around the 0 line, suggesting that the variances of the error terms are equal (Pennsylvania State University, 2018).

Normal Q-Q Plots and Shapiro-Wilk test

A Q-Q plot is used to assess whether the residuals are normally distributed. If the residuals are normally distributed, the points in the Q-Q plot should lie along a straight line. Large deviations from the line indicate departures from normality, such as skewness or heavy tails, and a $p - value > 0.05$ indicates that the data are normally distributed (Belsley et al., 2005).

Leverage and Influencial Plots

Leverage is a measure of the effect to of a particular observation on the regression predictions. An observation is influential if its removal significantly changes the regression estimates (Chatterjee & Hadi, 2015). Cook's Distance measures the influence of each data point on the fitted values.

Variance Inflation Factor

The Variance Inflation Factor is used to check for multicollinearity. High VIF values indicate a high correlation between the predictor variables, which can lead to unreliable estimates of regression coefficients (Fox, 2015). A VIF value greater than 10 indicates the presence of multicollinearity.

Remedial Measures for Assumption Violations

When assumptions in linear regression are violated, it's crucial to take action. Non-normality, heteroscedasticity, and multicollinearity can mess up your model's accuracy and reliability. For non-normal residuals, the data is transformed. Common transformations to address non-normality include logarithmic (log), square root, and Box-Cox transformations Logarithmic transformations are suitable when the residuals exhibit right-skewness. Square root transformations are appropriate for moderately right-skewed residuals. Box-Cox transformations provide a more flexible approach by estimating an optimal power transformation parameter (λ) (R. M. A. V., 2018).

Generalized Additive Model

A generalized additive model is a generalized linear model with a linear predictor involving a sum of smooth functions of covariates (Wood, 2017). Generalized Additive Models are an extension of the generalized linear model that allows for non-linear relationships between the dependent and independent variables (Hastie & Tibshirani, 1986).GAM models the response variable as a sum of smooth functions of the predictors (Grego, 2006). The model is in the form:

$$g(\mathbb{E}[Y \mid X_1, X_2, \ldots, X_p]) = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p)$$

K-Fold Cross Validation

One effective method for evaluating model performance and mitigating overfitting is K-Fold Cross Validation. Rather than relying on a single train-test split, K-Fold Cross Validation partitions the data into multiple subsets to ensure a more robust estimate of the model's predictive performance. The procedure is as follows:

- The dataset is randomly divided into K equal-sized folds.
- The model is trained on $K - 1$ folds and validated on the remaining fold.

- This process is repeated K times, with each fold used once as the validation set.
- The final performance metric is obtained by averaging the results across all K iterations.

For this analysis, we selected $K = 2$, which offers a reasonable trade-off between computational efficiency and variance reduction(James et al., 2013).

Model Evaluation

The performance of the model was assessed using two key metrics: the root mean square error (RMSE) and the statistic R^2 (R-squared). The RMSE measures the average magnitude of the residuals, providing insight into the model's prediction accuracy. In contrast, R^2 quantifies the proportion of variance in the dependent variable that is explained by the model. These metrics jointly provide a comprehensive understanding of both predictive accuracy and model fit.

Root Mean Squared Error (RMSE)

The Root Mean Squared Error (RMSE) quantifies the average difference between the predicted and observed values. It is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

A lower RMSE indicates better predictive accuracy of the model.

Coefficient of Determination (R^2)

The coefficient of determination, denoted R^2, measures the proportion of variance in the observed data that is explained by the model. It is given by:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

An R^2 value close to 1 indicates that the model explains a large proportion of the variance in the dependent variable, while a value closer to 0 suggests limited explanatory power.

0.0.4 Results

Exploratory Analysis

From Table 1, the analysis of $n = 522$ residential properties revealed a mean sales price of $\mu = \$277,894.1$ (SD $= \sigma = \$137,923.4$) and a median of $\$229,900$. The average finished square footage was $\mu_{sqft} =$ 2260.6 sq. ft. ($\sigma_{sqft} = 711.1$). The modal number of bedrooms was 3 (38.7%) and bathrooms was 3 (33.5%). Most properties featured 2-car garages (67.6%) and air conditioning (83.1%), with a mean lot size of $\mu_{lot} = 24369.7$ sq. ft. ($\sigma_{lot} = 11684.1$). The majority of properties were constructed between 1946 and 1985 (73.8%) and were of medium quality (55.6%). From Figure 1, Sale prices exhibit a right-skewed distribution, concentrated between $\$150,000$ and $\$300,000$ with high-end outliers. Year built shows a bimodal, right-truncated distribution peaking in the 1950s and late 1970s. Lot size and finished square footage are also right-skewed, with most properties ranging from $15,000$ to $30,000$ sq ft and $1,500$ to $2,500$ sq ft, respectively, and notable upper outliers. Boxplot analysis indicates that air conditioning is associated with a higher median sales price ($\sim \$250,000$) compared to its absence ($\sim \$175,000$). Median sale price shows a positive trend with increasing garage capacity: 2-car ($\sim \$250,000$), 3-car ($\sim \$400,000$), and $>$ 3-car ($\sim> \$500,000$). Homes with pools have a higher median price ($\sim \$350,000$) than those without ($\sim \$275,000$). Notably, quality level strongly influences price, with high-quality homes ($\sim \$500,000$) exceeding medium ($\sim \$275,000$) and low ($\sim \$200,000$) quality properties. Median sales prices are higher for homes not adjacent to highways ($\sim \$250,000$) compared to adjacent homes ($\sim \$230,000$). Architectural style influences median price: 7–11 elements ($\sim \$350,000$) command higher values than styles 1–3 and 4–6 ($\sim \$240,000$ and $\sim \$260,000$, respectively). Median price increases with

bedroom count: 5–7 bedrooms ($\sim$ \$400,000–\$450,000) versus 1–3 bedrooms ($\sim$ \$200,000–\$275,000). Similarly, higher bathroom counts correlate with increased median prices: $\geq$5 bathrooms ($\sim>$ \$500,000) compared to 1–2 bathrooms ($\sim$ \$175,000–\$300,000). From Figure 2, Strong positive correlations exist between sales price (P) and finished square footage (ft^2) ($\rho \approx 0.82$), moderately with bathrooms (B) ($\rho \approx 0.68$) and garage size (G) ($\rho \approx 0.58$). Lot size (L) shows weak correlations ($\rho < 0.3$) with all variables. High intercorrelation between ft^2 and B ($\rho \approx 0.76$) signals potential multicollinearity.

Regression Model Fit

Tbale 2 presents the linear regression model, several predictors significantly influence sales price ($p < 0.05$). *Finished square footage* positively predicts price ($\beta = 101.8$, $p < 0.001$), indicating that each additional square foot increases the expected sales price by approximately \$102, holding other variables constant. *Medium-quality* homes ($\beta = -37{,}842$, $p < 0.001$) and *low-quality* homes ($\beta = -82{,}285$, $p < 0.001$) are significantly cheaper than high-quality ones, reflecting reductions in expected price by about \$37.8K and \$82.3K, respectively. *Number of bathrooms* also shows a positive effect ($\beta = 10{,}482$, $p = 0.015$), where each additional bathroom increases the expected price by around \$10.5K. Similarly, *garage size* has a significant impact ($\beta = 10{,}558$, $p = 0.035$), implying an approximate \$10.6K increase per additional garage space.

Table 3 & 8 presents the linear model has a coefficient of determination ($R^2 = 0.8287$), indicating the proportion of variance explained by the model, and a standard error ($\sigma = 58034.88$), representing the prediction error. Model fit and complexity are captured by the log-likelihood (logLik $= -6457.24$), Akaike Information Criterion (AIC $= 12952.48$), and Bayesian Information Criterion (BIC $= 13033.38$). The log-transformed model shows improved explanatory power with $R^2 = 0.8342$ and reduced standard error ($\sigma = 0.01826$). Model fit statistics also improve, with logLik $= 1358.07$, AIC $= -2678.13$, and BIC $= -2597.24$.

Model Diagnosis for Regression Model

Table 4 evaluates the regression residuals' adherence to key assumptions, employing a significance level of $\alpha = 0.05$. The Shapiro-Wilk test yields a test statistic of ($W = 0.9374$, p-value < 0.0001), leading to the rejection of the null hypothesis of normality. The Breusch-Pagan test returns a test statistic of ($BP = 147.253$, p value < 0.0001, thus rejecting the null hypothesis of homoscedasticity and confirming the presence of heteroscedasticity.

Table 5 presents the Variance Inflation Factor (VIF) values indicate multicollinearity among predictors. Finished square feet ($VIF = 3.87$)and number of bedrooms ($VIF = 4.54$) exhibit collinearity, while other variables, such as garage size $VIF = 1.03$, and number of bathrooms ($VIF = 1.78$), show minimal multicollinearity. From Figure 3, the residual plot reveals heteroscedasticity, as residual variance increases with fitted values, and non-linearity, indicated by the curved LOESS line's deviation. These patterns suggest that the linear model's assumptions are violated. The Q-Q plot reveals mild right skewness, with residuals deviating from the 45-degree line in both tails, indicating non-normality. However, the histogram appears bell-shaped, approximating symmetry. These plots suggest violations of normality assumptions, particularly in the distribution's tails.

From Figure 4, the Cook's Distance plot reveals a few data points with higher influence. While most observations fall below the influence threshold. All values above the red dotted line are leverage values which influence the regression estimates; the number of influential observations from the dataset are 44. From Table 6, Observations with Cook's D values above $0.05(0.07497, 0.07454)$ and large standardised residuals $(-4.03, 4.72)$ suggest potential influential outliers affecting model stability and accuracy.

Model Evaluation for the Log Model and the Regression Model

Table 9 compares the original scale (linear) and log-transformed models using cross-validation. The log-transformed model shows a slight improvement in R^2 (0.8126 vs. 0.8054) and a substantial reduction in MSE (3.65×10^{-4} vs. 3.76×10^9) and RMSE (0.0191 vs. 61330.59). The log-transformed model also exhibits lower standard deviations for both MSE and R^2, indicating more stable performance. The log-transformed model has lower AIC (-2678.13 vs. 12952.48) and BIC (-2597.24 vs. 13033.38), further supporting its superior fit.
From 5, the optimal Box-Cox transformation parameter $\lambda = -0.195$, we apply log transformation on the response variable.

Model Diagnosis After Transformation

Table 10 presents the Shapiro-Wilk test for normality of the log model residuals yields a ($W = 0.9915, p = 0.0041$), indicating a significant departure from normality. This suggests that the log transformation did not fully normalise the residuals. The Q-Q plot shows minor deviations at the tails but generally aligns with the theoretical line, indicating approximate normality of residuals. The residuals vs fitted plot shows some non-linearity and slight heteroscedasticity at higher fitted values, confirming that the log transformation improves model assumptions but does not fully resolve all issues. The residuals vs fitted plot shows slight non-linearity and mild heteroscedasticity but is generally centered around zero. The histogram of residuals appears approximately symmetric and bell-shaped, supporting near-normality. The predicted vs actual plot displays a strong linear pattern along the diagonal, indicating good predictive performance of the log-transformed model.

Rank of Predictors for the Log Model

Table 11 Predictor importance, assessed by Root Mean Square Error (RMSE), shows finished square footage as most influential ($RMSE = 0.0206$), followed by quality ($RMSE = 0.0227$) and bathrooms ($RMSE = 0.0249$). Variables like lot size ($RMSE = 0.0363$) and highway adjacency ($RMSE = 0.0373$) have marginal impact, indicating that size and quality are key determinants of price variation.

Generalised Additive Model Fit

Table 12 , predictorS importance of GAM is ranked by RMSE. *Finished square footage* has the lowest RMSE (0.0194), indicating highest importance. *Quality* (RMSE = 0.0229) and *number of bathrooms* (RMSE = 0.0240) are also strong predictors. Other predictors, ranked by increasing RMSE, are: *Year Built* (0.0289), *Garage Size* (0.0306), *Number of Bedrooms* (0.0327), *Style* (0.0335), *Air Conditioning* (0.0350), *Lot Size* (0.0353), *Pool* (0.0369), and *Adjacent to Highway* (0.0373), the least important.

Model Evaluation of Log Model and GAM

Table 13 presents a comparison of the log-transformed linear model (LM) and the Generalized Additive Model (GAM) using Root Mean Square Error (RMSE) shows that GAM generally outperforms LM. The greatest RMSE reduction in GAM is observed for *finished square footage* (LM RMSE = 0.02409, GAM RMSE = 0.0194, ΔRMSE = -0.0010), followed by *lot size* (ΔRMSE = -0.0010) and *number of bathrooms* (ΔRMSE = -0.0014). Conversely, LM has slightly lower RMSE for *garage size*, *pool*, *air conditioning* and *quality*, but the differences are minimal.

0.0.5 Discussion

This study's comparative analysis of linear regression and generalized additive models (GAMs) for real estate price prediction yields critical insights into modern valuation methodologies.

The linear regression model's strong explanatory power ($R = 0.8287$) confirms its utility as a baseline valuation tool, particularly for establishing clear parametric relationships between key property characteristics and sales prices. The significant coefficients for square footage and construction quality align with established hedonic pricing theory (Rosen, 1974), validating conventional wisdom about value determinants. However, the diagnostic results reveal substantial violations of linear modeling assumptions, particularly regarding residual distribution and variance stability. These findings corroborate growing concerns in the literature about applying rigid linear frameworks to inherently non-linear real estate markets (Selim, 2009; Brunauer et al., 2010).

The log transformation's partial success in improving model fit while failing to fully normalize residuals presents an important methodological insight. This outcome suggests that while variable transformations can mitigate some linear model limitations, they may be insufficient for addressing the structural complexities present in real estate data. The persistence of multicollinearity among key predictors further underscores the challenges of applying traditional regression techniques to highly correlated property features.

The GAM's superior performance, particularly for spatial and quality-related variables, provides compelling evidence for adopting more flexible modeling approaches in valuation practice. The model's ability to capture non-linear relationships through smooth functions (Wood, 2017) proved especially

valuable for characteristics like lot size and construction year, where price effects are known to be non-monotonic (Yang, 2025). The cross-validation results demonstrate that GAMs can achieve meaningful improvements in predictive accuracy without sacrificing statistical rigor.

0.0.6 Conclusion

This comparative study between linear regression and generalized additive models (GAMs) in real estate price prediction yields significant insights for both researchers and practitioners. While linear regression maintains value through its interpretability and established methodology, its inability to fully capture non-linear relationships and heteroscedastic residuals limits its predictive accuracy in complex housing markets. The GAM framework demonstrates superior performance, particularly in modeling key non-linear predictors like square footage and construction year, while maintaining statistical rigor. However, this enhanced predictive capability comes at the cost of reduced model transparency.

The findings suggest that optimal real estate valuation may require a tiered modeling approach: employing linear models for initial exploratory analysis and regulatory applications where interpretability is paramount, while reserving GAMs for high-stakes predictive tasks.

0.0.7 Recommendation

For Policy Makers

1. Adopt Generalized Additive Models (GAMs) for predictive modeling of real estate sales.
2. Maintain linear models for explanatory analysis due to their interpretability.
3. Promote training and capacity building by incorporating GAM methodologies into valuation certification programs.

Future Research

1. Develop more interpretable non-linear modeling techniques.
2. Investigate ensemble methods that combine multiple modeling approaches.
3. Explore spatial-temporal extensions to better capture market dynamics.
4. Establish standardized benchmarking frameworks for model comparison.

0.0.8 References

ChatGPT was used to correct my codes and some my wordings

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0.0.9 Appendix A

Regression Model
The general form of a regression model:

$$y = X\beta + \varepsilon$$

Where,

- y is the response variable Sales price, which is an $n \times 1$.
- X is an $n \times p$ matrix corresponding to the predictor variables : Finished Square Feet, Number of Bedrooms, Number of Bathrooms, Air Conditioning, Garage Size, Pool, Year Built, Quality, Style, Lot Size and Adjacent to Highway.
- β is an $n \times 1$ vector containing of the regression coefficients of the predictor variables.
- ε is an $n \times 1$ vector of the errors.

Parameter Estimates
The parameters are estimated using **OLS and MLE**. To find the vector $\hat{\beta} = (b_1, b_2, \ldots, b_k)' \in \mathcal{B}$ that minimizes the sum of squared residuals.

The vector of residuals ϵ is

$$\epsilon = Y - X\beta$$

The sum of squared residuals is:

$$S(\beta) = \sum_{i=1}^{n} \epsilon_i^2 = \epsilon'\epsilon = (Y - X\beta)'(Y - X\beta)$$

$$S(\beta) = Y'Y + \beta' X' X\beta - 2\beta' X' Y$$

$$\frac{\partial S(\beta)}{\partial \beta} = 2X'X\beta - 2X'Y$$

$$2X'X\beta - 2X'Y = 0$$

$$\hat{\beta} = (X'X)^{-1}X'Y$$

where,

$$\hat{\beta} = (X'X)^{-1}X'Y$$

is the parameter estimates using the OLS

To the estimates using the MLE approach for β and σ^2, we assume

$$\epsilon_i \sim N(0, 1)$$

i.e

$$f(\epsilon_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\epsilon_i^2}{2\sigma^2}\right)$$

$$L(\beta, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - x_i\beta)^2}{2\sigma^2}\right)$$

$$\ln L(\beta, \sigma^2) = \sum_{i=1}^{n} \left[-\frac{1}{2}\ln(2\pi\sigma^2) - \frac{(y_i - x_i\beta)^2}{2\sigma^2}\right]$$

$$\ln L(\beta, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - x_i\beta)^2$$

First, we take the derivative of the log-likelihood concerning β:

$$\frac{\partial}{\partial\beta}\ln L(\beta, \sigma^2) = -\frac{1}{2\sigma^2} \cdot \frac{\partial}{\partial\beta}\sum_{i=1}^{n}(y_i - x_i\beta)^2$$

$$\frac{\partial}{\partial\beta}(y_i - x_i\beta)^2 = -2x_i(y_i - x_i\beta)$$

$$\frac{\partial}{\partial\beta}\ln L(\beta, \sigma^2) = \frac{1}{\sigma^2}\sum_{i=1}^{n} x_i(y_i - x_i\beta)$$

$$\sum_{i=1}^{n} x_i(y_i - x_i\beta) = 0$$

$$X'(Y - X\beta) = 0$$

$$\hat{\beta} = (X'X)^{-1}X'Y$$

Next, we take the derivative of the log-likelihood for σ^2:

$$\frac{\partial}{\partial \sigma^2} \ln L(\beta, \sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^{n} (y_i - x_i\beta)^2$$

$$-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^{n} (y_i - x_i\beta)^2 = 0$$

$$n\sigma^2 = \sum_{i=1}^{n} (y_i - x_i\hat{\beta})^2$$

Thus, the MLE for σ^2 is:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (y_i - x_i\hat{\beta})^2$$

The Maximum Likelihood Estimates for β and σ^2 are:

$$\hat{\beta} = (X'X)^{-1}X'Y$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (y_i - x_i\hat{\beta})^2$$

Leverage and Influencial Plots
An observation can be considered as an outlier if its leverage substantially exceeds the mean leverage value $\frac{p}{n}$. The leverage satisfy $0 \leq h_{ii} \leq 1$ and

$$\sum_{i=1}^{n} h_{ii} = \text{tr}(H) = \text{tr}(X(X'X)^{-1}X') = p$$

where p is the number of parameters.
Cook's Distance measures the influence of each data point on the fitted values:

$$D_i = \frac{e_i^2}{k \cdot \hat{\sigma}^2} \cdot \frac{h_{ii}}{(1 - h_{ii})^2}$$

,

$$\text{If } D_i > 1, \text{ then the } i\text{-th observation is considered influential.}$$

General Additive Model
The general form of the GAM is:

$$g(\mathbb{E}[Y \mid X_1, X_2, \ldots, X_p]) = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p)$$

where:

- $g(\cdot)$ is the link function that relates the expected value of the response variable Y to the additive predictor.
- β_0 is the intercept term.
- $f_j(\cdot)$ for $j = 1, 2, \ldots, p$ are smooth functions of the predictor variables X_j
- X_j are the response variables.

Parameter Estimation in GAM

$$\hat{\beta}_0, \hat{f}_1, \ldots, \hat{f}_p = \arg \max_{\beta_0, f_1, \ldots, f_p} \left\{ \ell(Y; \beta_0, f_1, \ldots, f_p) - \frac{1}{2} \sum_{j=1}^{p} \lambda_j \int \left(f_j''(x) \right)^2 dx \right\}$$

Where,

- $\ell(Y; \beta_0, f_1, \ldots, f_p)$ is the log-likelihood of the response variable Y given the model.
- λ_j is a smoothing parameter controlling the complexity of the smooth function f_j.
- $\int \left(f_j''(x)\right)^2 \, dx$ is the roughness penalty for function f_j, penalizing excessive wiggliness.

Root Mean Squared Error (RMSE)
The Root Mean Squared Error (RMSE) quantifies the average difference between the predicted and observed values. It is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

where:

- y_i denotes the observed value for observation i,
- $\hat{y}_i$ denotes the predicted value for observation i,
- n is the total number of observations.

Let $e_i = y_i - \hat{y}_i$ represent the residual for observation i, then, the Mean Squared Error (MSE) is:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n} e_i^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

A lower RMSE implies better fit.

Coefficient of Determination (R^2)
The coefficient of determination, denoted R^2, measures the proportion of variance in the observed data that is explained by the model. It is given by:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

where:

- $\bar{y}$ is the mean of the observed values $y_1, y_2, \ldots, y_n$,
- The numerator represents the residual sum of squares (unexplained variance),
- The denominator represents the total sum of squares (total variance).

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$$
$$\Rightarrow \text{SST} = \text{SSR} + \text{SSE}$$

Then:

$$R^2 = \frac{\text{SSE}}{\text{SST}} = 1 - \frac{\text{SSR}}{\text{SST}}$$

- $R^2 = 1$: The model explains all the variability in the data.
- $R^2 = 0$: The model explains none of the variability.
- $R^2 < 0$: The model fits worse than a horizontal line at $\bar{y}$.

Remedial Measures for Violated Regression Assumptions

Logarithmic Transformation
Let $Y \sim \text{Log-Normal}(\mu, \sigma^2)$, so that $\log(Y) \sim N(\mu, \sigma^2)$. Define the transformed variable:

$$Z = \log(Y)$$

$$Z = \log(Y) = \beta_0 + \beta_1 X + \varepsilon$$

$$\text{Var}[\log(Y)] \approx \left(\frac{d \log(Y)}{dY} \bigg|_{Y=\mu} \right)^2 \cdot \text{Var}(Y) = \left(\frac{1}{\mu} \right)^2 \cdot \sigma^2 = \frac{\sigma^2}{\mu^2}$$

Hence, the log transformation is appropriate when $\text{Var}(Y) \propto \mu^2$, especially when Y is right-skewed.

Square Root Transformation

If the variance of Y increases linearly with the mean, i.e., $\text{Var}(Y) \propto \mu$, a square root transformation may stabilize variance:

$$Z = \sqrt{Y}$$

$$\text{Var}(\sqrt{Y}) \approx \left(\frac{d\sqrt{Y}}{dY} \bigg|_{Y=\mu} \right)^2 \cdot \text{Var}(Y) = \left(\frac{1}{2\sqrt{\mu}} \right)^2 \cdot \sigma^2 = \frac{\sigma^2}{4\mu}$$

This transformation is particularly suitable for moderately right-skewed data.

Box-Cox Transformation
The Box-Cox transformation generalizes these two approaches by estimating the optimal transformation:

$$Y^{(\lambda)} = \begin{cases} \frac{Y^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log(Y), & \lambda = 0 \end{cases}$$

The transformed model becomes:

$$Y_i^{(\lambda)} = \beta_0 + \beta_1 X_i + \varepsilon_i$$

The maximum likelihood estimation (MLE) of λ involves maximizing the log-likelihood:

$$\log L(\lambda) = -\frac{n}{2} \log \left(\frac{1}{n} \sum_{i=1}^{n} \left(Y_i^{(\lambda)} - \hat{Y}_i^{(\lambda)} \right)^2 \right) + (\lambda - 1) \sum_{i=1}^{n} \log(Y_i)$$

- When $\lambda = 1$, no transformation is applied: $Y^{(\lambda)} = Y$.
- When $\lambda = 0$, the transformation corresponds to the natural logarithm: $Y^{(0)} = \log(Y)$.
- When $\lambda = 0.5$, the transformation is equivalent to the square root: $Y^{(0.5)} = \sqrt{Y}$.

0.0.10 Appendix B

Table 1: Descriptive Statistics of Residential Property Features

Variable	N(%)
__Sales price__	277,894.1 (137,923.4), 229,900.0 (180,000.0, 335,000.0)
__Finished square feet__	2,260.6 (711.1), 2,061.0 (1,701.0, 2,638.0)
__Number of bedrooms__	
0	1.0 (0.2%)
1	9.0 (1.7%)
2	64.0 (12.3%)
3	202.0 (38.7%)
4	179.0 (34.3%)
5	52.0 (10.0%)
6	12.0 (2.3%)
7	3.0 (0.6%)
__Number of bathrooms__	
0	1.0 (0.2%)
1	71.0 (13.6%)
2	171.0 (32.8%)
3	175.0 (33.5%)
4	84.0 (16.1%)
5	17.0 (3.3%)
6	1.0 (0.2%)
7	2.0 (0.4%)
__Air conditioning__	434.0 (83.1%)
__Garage size__	
0	7.0 (1.3%)
1	52.0 (10.0%)
2	353.0 (67.6%)
3	106.0 (20.3%)
4	2.0 (0.4%)
5	1.0 (0.2%)
7	1.0 (0.2%)
__Pool__	36.0 (6.9%)
__Year Built__	
1885-1905	2.0 (0.4%)
1906-1925	10.0 (1.9%)
1926-1945	27.0 (5.2%)
1946-1965	207.0 (39.7%)
1966-1985	178.0 (34.1%)
1986-1998	98.0 (18.8%)
__Quality__	
High Quality	68.0 (13.0%)
Medium Quality	290.0 (55.6%)
Low Quality	164.0 (31.4%)
__Style__	
1-3	336.0 (64.4%)
4-6	47.0 (9.0%)
7-11	139.0 (26.6%)
__Lot size__	24,369.7 (11,684.1), 22,200.0 (17,159.0, 26,790.0)
__Adjacent to highway__	11.0 (2.1%)

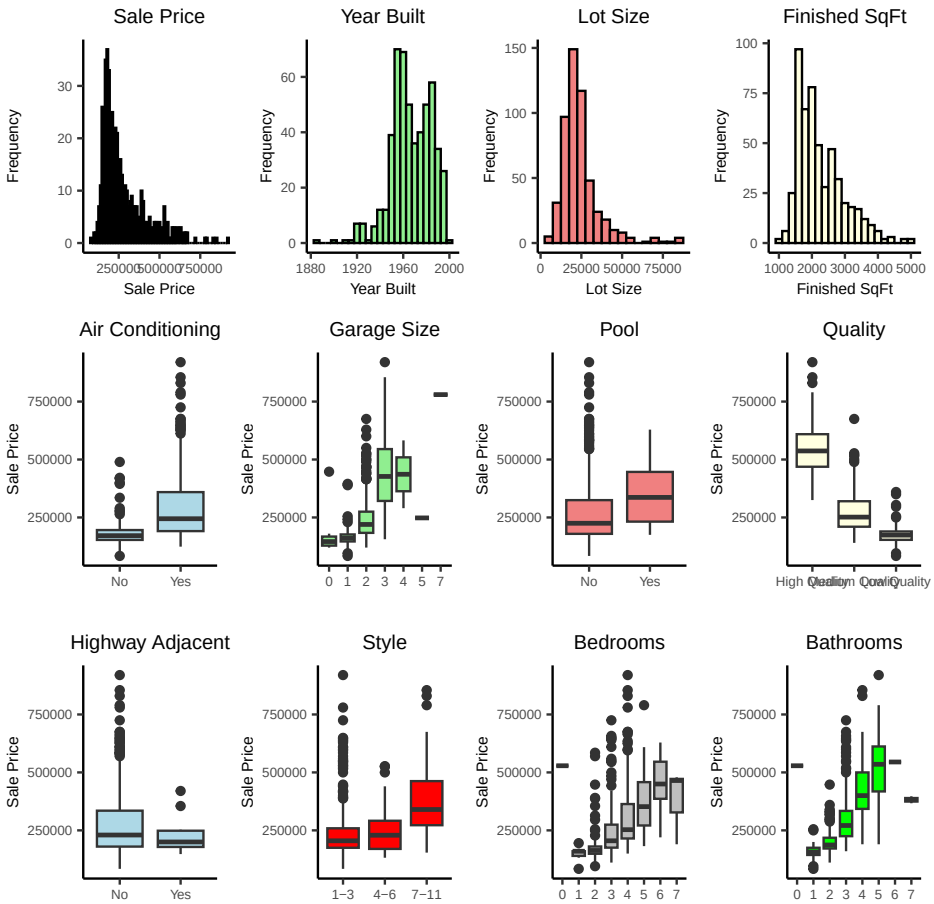

Figure 1: Histograms of the Numeric Variables and Box Plots of Sale Price by Categorical Variables

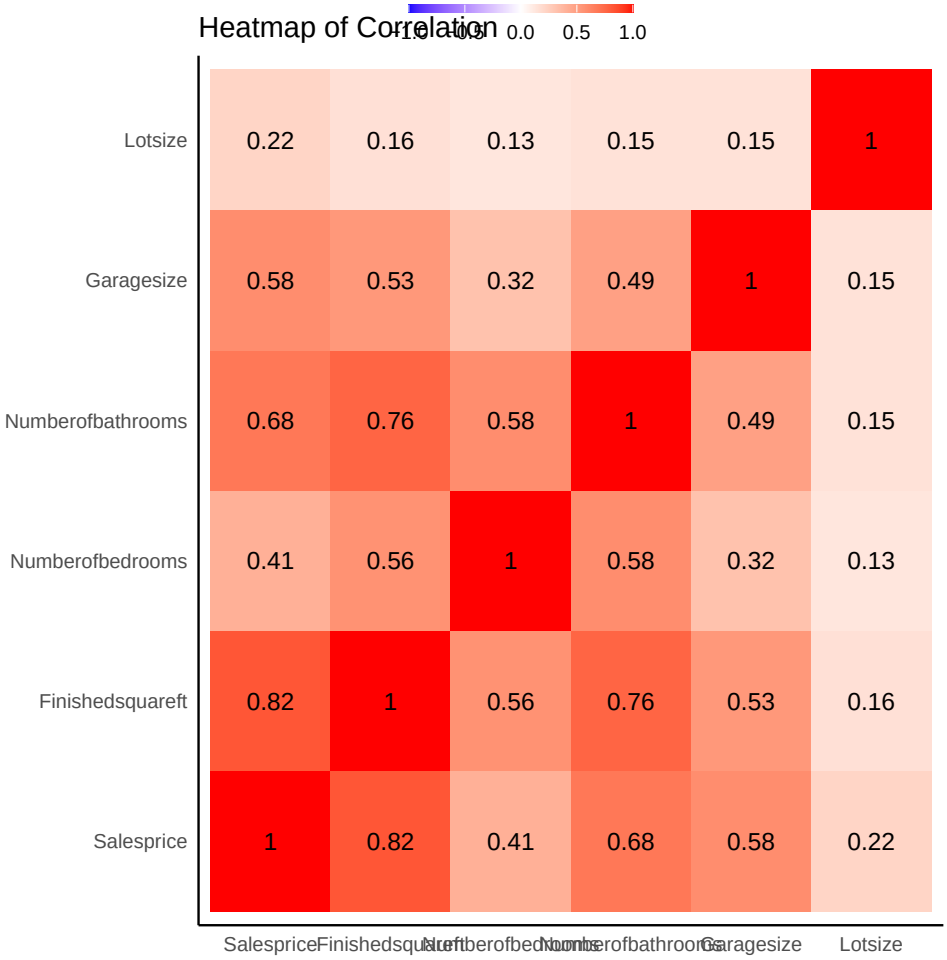

Figure 2: A Correlation Matrix of the Numeric Variables

Table 2: Parameter Estimates and P-values of the Linear Regression Model

term	estimate	std.error	statistic	p.value
(Intercept)	32106.306	47800.054	0.672	0.502
QualityMedium Quality	-128443.865	10648.226	-12.062	0.000
QualityLow Quality	-136135.216	14309.646	-9.514	0.000
Finishedsquareft	101.794	7.617	13.364	0.000
Numberofbedrooms	-3950.781	3344.968	-1.181	0.238
AdjacenttohighwayYes	-29142.737	17914.859	-1.627	0.104
Numberofbathrooms	10481.959	4295.836	2.440	0.015
AirconditioningYes	7082.573	7983.991	0.887	0.375
Garagesize	10558.524	5000.982	2.111	0.035
PoolYes	13242.137	10535.149	1.257	0.209
Style4-6	-10673.308	10527.614	-1.014	0.311
Style7-11	-40681.123	8148.301	-4.993	0.000
Lotsize	1.277	0.237	5.382	0.000
YearBuilt1906-1925	27440.826	45254.727	0.606	0.545
YearBuilt1926-1945	46954.528	42986.719	1.092	0.275
YearBuilt1946-1965	52736.832	42339.109	1.246	0.213
YearBuilt1966-1985	69760.589	42827.560	1.629	0.104
YearBuilt1986-1998	109034.414	43242.713	2.521	0.012

Table 3: Summary Statistics for the Linear Regression Model

	Statistic	Value
r.squared	r.squared	8.287000e-01
adj.r.squared	adj.r.squared	8.229000e-01
sigma	sigma	5.803488e+04
statistic	statistic	1.434488e+02
p.value	p.value	0.000000e+00
df	df	1.700000e+01
logLik	logLik	-6.457240e+03
AIC	AIC	1.295248e+04
BIC	BIC	1.303338e+04
deviance	deviance	1.697496e+12
df.residual	df.residual	5.040000e+02
nobs	nobs	5.220000e+02

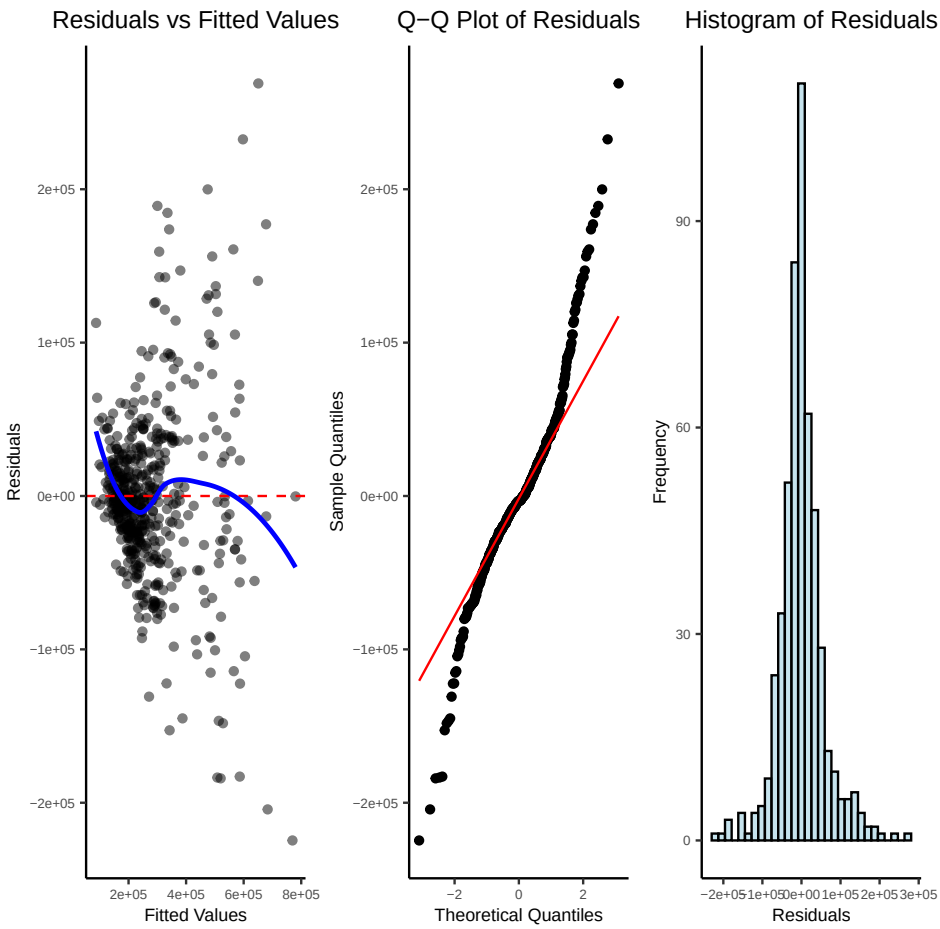

Figure 3: A Q-Q, Histogram and Residuals vs. Fitted values Plot

Table 4: Diagnostic Tests: Breusch-Pagan and Shapiro-Wilk

Test	Test Statistic	P-value
Breusch-Pagan	147.2530	0
Shapiro-Wilk	0.9374	0

Table 5: Variance Inflation Factors (VIF) for Multicollinearity Assessment

	Predictor	GVIF	Df	Adjusted GVIF
Quality	Quality	3.868	2	1.402
Finishedsquareft	Finishedsquareft	4.538	1	2.130
Numberofbedrooms	Numberofbedrooms	1.781	1	1.334
Adjacenttohighway	Adjacenttohighway	1.026	1	1.013
Numberofbathrooms	Numberofbathrooms	3.233	1	1.798
Airconditioning	Airconditioning	1.385	1	1.177
Garagesize	Garagesize	1.655	1	1.286
Pool	Pool	1.105	1	1.051
Style	Style	2.471	2	1.254
Lotsize	Lotsize	1.190	1	1.091
YearBuilt	YearBuilt	3.301	5	1.127

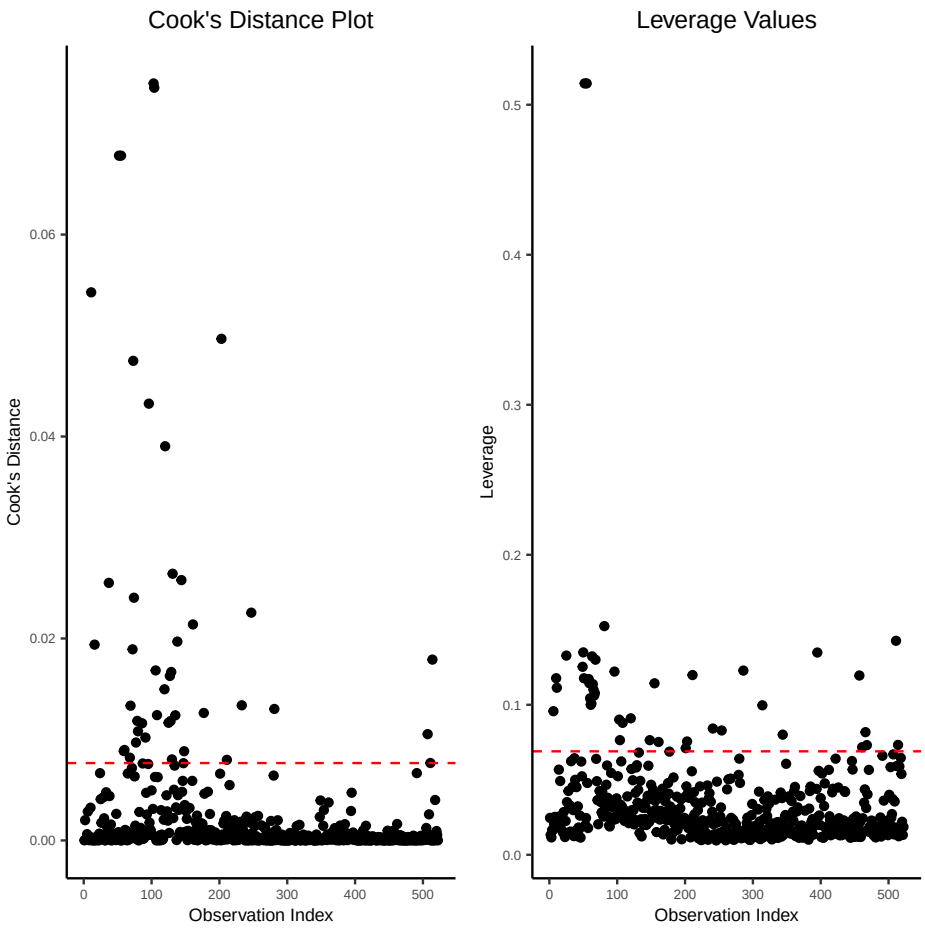

Figure 4: A Cook's Distance and Leverage Plot

Table 6: Top Influential Observations Based on Cook's Distance

Cook's D	Fitted	Residual	Std. Residual
0.07496623	683342.6	-204342.57	-3.6913
0.07453892	769587.3	-224587.35	-4.0268
0.06781212	116562.8	43437.22	1.0739
0.06781212	198437.2	-43437.22	-1.0739
0.05427633	342765.3	-152765.32	-2.7923
0.04967416	335409.5	184590.48	3.3081
0.04749535	651019.3	268980.74	4.7228
0.04325286	471377.8	128622.19	2.3655
0.03903503	513614.0	-146613.98	-2.6497
0.02641175	520106.9	-184106.86	-3.2432

```
[1] "Number of high leverage points: 44"
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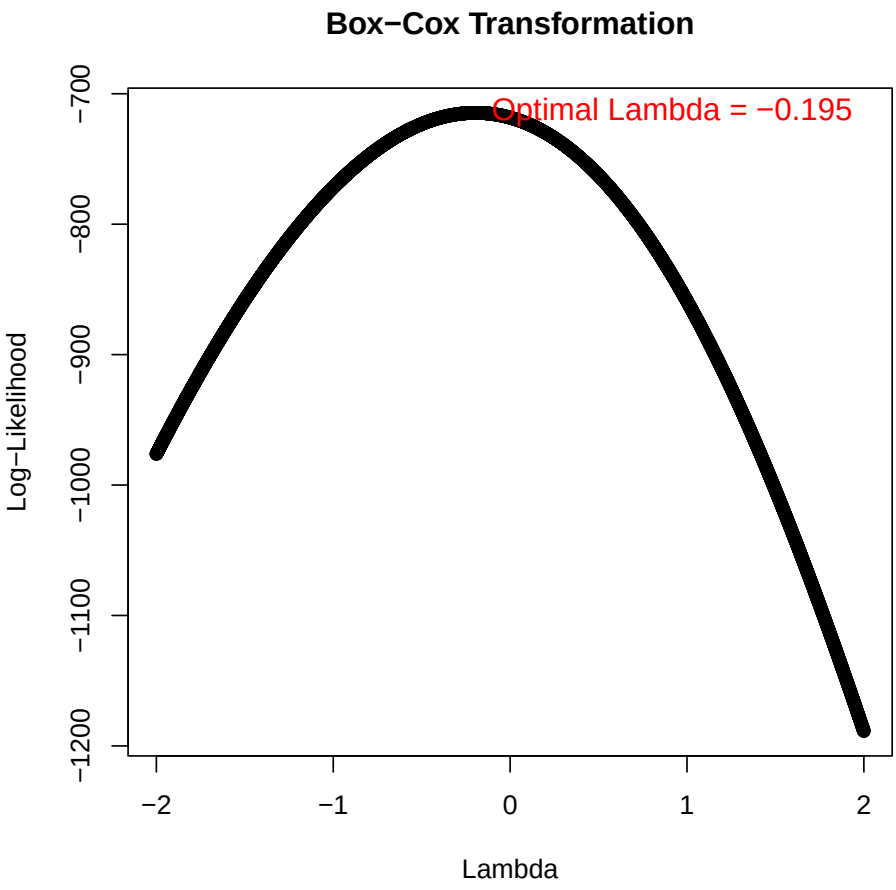

Figure 5: Box-Cox Log-Likelihood Plot Showing Optimal Lambda

Table 7: Log-Transformed Model Coefficients

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	4.5815	0.0128	358.8969	0.0000	4.5564	4.6066
QualityMedium Quality	-0.0201	0.0028	-7.0759	0.0000	-0.0257	-0.0145
QualityLow Quality	-0.0297	0.0038	-7.7592	0.0000	-0.0372	-0.0221
Finishedsquareft	0.0000	0.0000	12.1631	0.0000	0.0000	0.0000
Numberofbedrooms	0.0007	0.0009	0.8045	0.4215	-0.0010	0.0025
AdjacenttohighwayYes	-0.0057	0.0048	-1.1987	0.2312	-0.0151	0.0037
Numberofbathrooms	0.0040	0.0011	3.5280	0.0005	0.0018	0.0063
AirconditioningYes	0.0061	0.0021	2.8443	0.0046	0.0019	0.0103
Garagesize	0.0036	0.0013	2.6635	0.0080	0.0009	0.0062
PoolYes	0.0052	0.0028	1.8617	0.0632	-0.0003	0.0108
Style4-6	0.0010	0.0028	0.3634	0.7165	-0.0045	0.0065
Style7-11	-0.0081	0.0022	-3.7042	0.0002	-0.0123	-0.0038
Lotsize	0.0000	0.0000	6.4882	0.0000	0.0000	0.0000
YearBuilt1906-1925	0.0138	0.0121	1.1425	0.2538	-0.0099	0.0376
YearBuilt1926-1945	0.0148	0.0115	1.2852	0.1993	-0.0078	0.0373
YearBuilt1946-1965	0.0171	0.0113	1.5104	0.1316	-0.0051	0.0393
YearBuilt1966-1985	0.0240	0.0114	2.0975	0.0364	0.0015	0.0465
YearBuilt1986-1998	0.0319	0.0115	2.7591	0.0060	0.0092	0.0546

Table 8: Log Model Summary

	Statistic	Value
r.squared	r.squared	0.8339
adj.r.squared	adj.r.squared	0.8283
sigma	sigma	0.0155
statistic	statistic	148.8060
p.value	p.value	0.0000
df	df	17.0000
logLik	logLik	1443.6458
AIC	AIC	-2849.2915
BIC	BIC	-2768.3959
deviance	deviance	0.1211
df.residual	df.residual	504.0000
nobs	nobs	522.0000

```
k  power.n actual.n
2      220      220
```

Table 9: Model Evaluation for the Log Model and the Regression Model

Metric	Original (Sales price scale)	Log-transformed (LogSales price scale)
MSE	3.761441e+09	0.0003
RMSE	6.133059e+04	0.0162
R_squared	8.054000e-01	0.8122
MSE_SD	2.549687e+08	0.0000
R_squared_SD	5.210000e-02	0.0154

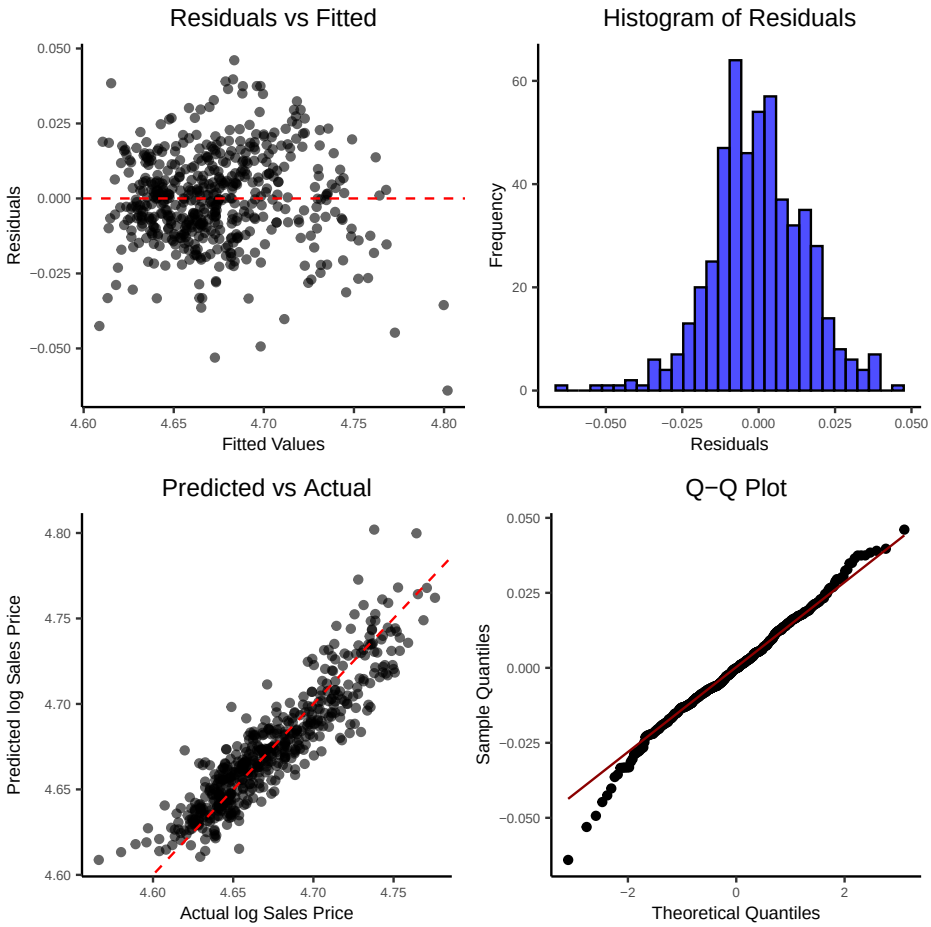

Figure 6: Model Diagnosis Plot for the Log Model

Table 10: Shapiro-Wilk Test for Normality of Log Model Residuals

	Test	W Statistic	p-value
W	Shapiro-Wilk (Log Model)	0.9914	0.0041

Table 11: Rank of the Predictors for the Log Model (RMSE)

	Predictor	RMSE
2	Finishedsquareft	0.0206
1	Quality	0.0228
5	Numberofbathrooms	0.0247
11	YearBuilt	0.0293
7	Garagesize	0.0314
3	Numberofbedrooms	0.0325
8	Style	0.0338
6	Airconditioning	0.0351
10	Lotsize	0.0365
9	Pool	0.0369
4	Adjacenttohighway	0.0377

Table 12: Ranking the Predictors Using the GAM (RMSE)

	Predictor	RMSE
2	Finishedsquareft	0.0196
1	Quality	0.0228
5	Numberofbathrooms	0.0239
11	YearBuilt	0.0293
7	Garagesize	0.0301
3	Numberofbedrooms	0.0326
9	Style	0.0337
6	Airconditioning	0.0351
10	Lotsize	0.0356
8	Pool	0.0371
4	Adjacenttohighway	0.0377

Table 13: Comparison of RMSE for Log and GAM Models by the Predictors

	Predictor	LM RMSE	GAM RMSE	RMSE Difference	Better Model
10	Lotsize	0.0363	0.0353	-1e-03	GAM
2	Finishedsquareft	0.0204	0.0194	-1e-03	GAM
5	Numberofbathrooms	0.0246	0.0238	-8e-04	GAM
7	Garagesize	0.0300	0.0292	-7e-04	GAM
4	Adjacenttohighway	0.0371	0.0374	3e-04	LM
1	Quality	0.0225	0.0227	2e-04	LM
8	Pool	0.0368	0.0370	1e-04	LM
9	Style	0.0336	0.0335	-1e-04	GAM
11	YearBuilt	0.0290	0.0290	1e-04	LM
3	Numberofbedrooms	0.0324	0.0324	0e+00	LM
6	Airconditioning	0.0349	0.0349	0e+00	LM